THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Real Numbers** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 2310	Highest powers of $2 \times 3 \times 5$, $2 \times 3 \times 7$, $2 \times 3 \times 11$
2	**(a)** 60	LCM of 1 to 6
3	**(b)** 9	HCF of the differences from the remainder
4	**(b)** 1	$11^2 = 121$; for odd n, n^2 leaves remainder 1 on division by 8
5	**(b)** $\dfrac{\sqrt{12}}{\sqrt{3}}$	$\sqrt{12}/\sqrt{3} = 2$
6	**(a)** $2^3 \times 5^2 \times 3$	Zeros at the end = min(power of 2, power of 5)
7	**(d)** 144	Highest powers of $2^2 \times 3^2$, $2^4 \times 3$, $2^3 \times 3^2$
8	**(d)** 420	LCM of 1 to 7
9	**(a)** 24	HCF of the differences from the remainder
10	**(b)** 1	$13^2 = 169$; for odd n, n^2 leaves remainder 1 on division by 8

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Numbers are $12 \times 3 = 36$ and $12 \times 5 = 60$ (1); LCM $= 180$ (1).	1+1
12	Each remainder is short of the divisor by 5 (1); LCM$(12, 18) = 36$, number $= 36 - 5 = 31$ (1).	1+1
13	If $3\sqrt{7} = r$ were rational, $\sqrt{7} = r/3$ would be rational (1) - a contradiction (1).	1+1
14	Numbers are $15 \times 4 = 60$ and $15 \times 7 = 105$ (1); LCM $= 420$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	LCM$(18, 24, 36) = 72$ min (2); rounds $4, 3, 2$ (1).	2+1
16	Required number $=$ HCF$(1250, 9376)$ (1); HCF$(1250, 9376) = 2$ (2).	1+2
17	Take 7 common: $7(5 \times 6 + 1)$ (2); $5 \times 6 + 1 = 31 > 1$, so there are two factors greater than 1 (1); composite.	2+1
	OR: LCM$(20, 28, 35) = 140$ min (2); rounds $7, 5, 4$ (1).	2+1
18	Required number $=$ HCF$(1016, 1628)$ (1); HCF$(1016, 1628) = 4$ (2).	1+2
	OR: Take 8 common: $8(6 \times 7 + 1)$ (2); $6 \times 7 + 1 = 43 > 1$, so there are two factors greater than 1 (1); composite.	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Books per stack $=$ HCF$(96, 240, 336) = 48$ (3); stacks $= \dfrac{672}{48} = 14$ (2).	3+2
	OR: If $1/\sqrt{2} = r$ were rational then $\sqrt{2} = 1/r$ would be rational - contradiction (3); $\dfrac{3}{\sqrt{2}} = 3 \times \dfrac{1}{\sqrt{2}}$ is a non-zero rational times an irrational, hence irrational (2).	3+2
20	Books per stack $=$ HCF$(144, 192, 288) = 48$ (3); stacks $= \dfrac{624}{48} = 13$ (2).	3+2
	OR: If $1/\sqrt{3} = r$ were rational then $\sqrt{3} = 1/r$ would be rational - contradiction (3); $\dfrac{5}{\sqrt{3}} = 5 \times \dfrac{1}{\sqrt{3}}$ is a non-zero rational times an irrational, hence irrational (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Real Numbers** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $\frac{\sqrt{75}}{\sqrt{3}}$	$\sqrt{75}/\sqrt{3} = 5$
2	**(c)** $2^4 \times 5^3 \times 3$	Zeros at the end = min(power of 2, power of 5)
3	**(c)** 900	Highest powers of $3^2 \times 5$, $2^2 \times 3 \times 5$, 3×5^2
4	**(a)** 840	LCM of 1 to 8
5	**(a)** 20	HCF of the differences from the remainder
6	**(a)** 1	$15^2 = 225$; for odd n, n^2 leaves remainder 1 on division by 8
7	**(c)** $\frac{\sqrt{108}}{\sqrt{3}}$	$\sqrt{108}/\sqrt{3} = 6$
8	**(d)** $2^5 \times 5^4 \times 3$	Zeros at the end = min(power of 2, power of 5)
9	**(b)** 312	Highest powers of $2^2 \times 13$, $2 \times 3 \times 13$, $2^3 \times 13$
10	**(a)** 2520	LCM of 1 to 9

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Each remainder is short of the divisor by 4 (1); LCM$(15, 20) = 60$, number $= 60 - 4 = 56$ (1).	1+1
12	If $5\sqrt{2} = r$ were rational, $\sqrt{2} = r/5$ would be rational (1) - a contradiction (1).	1+1
13	Numbers are $9 \times 5 = 45$ and $9 \times 8 = 72$ (1); LCM $= 360$ (1).	1+1
14	Each remainder is short of the divisor by 6 (1); LCM$(10, 14) = 70$, number $= 70 - 6 = 64$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	LCM$(15, 40, 60) = 120$ min (2); rounds $8, 3, 2$ (1).	2+1
16	Required number $=$ HCF$(940, 1388)$ (1); HCF$(940, 1388) = 4$ (2).	1+2
17	Take 9 common: $9(7 \times 8 + 1)$ (2); $7 \times 8 + 1 = 57 > 1$, so there are two factors greater than 1 (1); composite.	2+1
	OR: LCM$(16, 24, 30) = 240$ min (2); rounds $15, 10, 8$ (1).	2+1
18	Required number $=$ HCF$(2034, 2450)$ (1); HCF$(2034, 2450) = 2$ (2).	1+2
	OR: Take 10 common: $10(8 \times 9 + 1)$ (2); $8 \times 9 + 1 = 73 > 1$, so there are two factors greater than 1 (1); composite.	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Books per stack $=$ HCF$(120, 180, 300) = 60$ (3); stacks $= \frac{600}{60} = 10$ (2).	3+2
	OR: If $1/\sqrt{5} = r$ were rational then $\sqrt{5} = 1/r$ would be rational - contradiction (3); $\frac{7}{\sqrt{5}} = 7 \times \frac{1}{\sqrt{5}}$ is a non-zero rational times an irrational, hence irrational (2).	3+2
20	Books per stack $=$ HCF$(168, 252, 420) = 84$ (3); stacks $= \frac{840}{84} = 10$ (2).	3+2
	OR: If $1/\sqrt{7} = r$ were rational then $\sqrt{7} = 1/r$ would be rational - contradiction (3); $\frac{4}{\sqrt{7}} = 4 \times \frac{1}{\sqrt{7}}$ is a non-zero rational times an irrational, hence irrational (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Real Numbers** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 28	HCF of the differences from the remainder
2	**(c)** 1	$17^2 = 289$; for odd n, n^2 leaves remainder 1 on division by 8
3	**(d)** $\frac{\sqrt{147}}{\sqrt{3}}$	$\sqrt{147}/\sqrt{3} = 7$
4	**(d)** $2^6 \times 5^5 \times 3$	Zeros at the end = min(power of 2, power of 5)
5	**(a)** 252	Highest powers of $2^2 \times 7$, $2 \times 3 \times 7$, $3^2 \times 7$
6	**(a)** 2520	LCM of 1 to 10
7	**(c)** 14	HCF of the differences from the remainder
8	**(d)** 1	$19^2 = 361$; for odd n, n^2 leaves remainder 1 on division by 8
9	**(a)** $\frac{\sqrt{300}}{\sqrt{3}}$	$\sqrt{300}/\sqrt{3} = 10$
10	**(c)** $2^7 \times 5^6 \times 3$	Zeros at the end = min(power of 2, power of 5)

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	If $4\sqrt{11} = r$ were rational, $\sqrt{11} = r/4$ would be rational (1) - a contradiction (1).	1+1
12	Numbers are $18 \times 2 = 36$ and $18 \times 7 = 126$ (1); LCM $= 252$ (1).	1+1
13	Each remainder is short of the divisor by 7 (1); LCM$(16, 24) = 48$, number $= 48 - 7 = 41$ (1).	1+1
14	If $6\sqrt{3} = r$ were rational, $\sqrt{3} = r/6$ would be rational (1) - a contradiction (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	LCM$(22, 33, 44) = 132$ min (2); rounds $6, 4, 3$ (1).	2+1
16	Required number $=$ HCF$(1812, 2412)$ (1); HCF$(1812, 2412) = 12$ (2).	1+2
17	Take 11 common: $11(9 \times 10 + 1)$ (2); $9 \times 10 + 1 = 91 > 1$, so there are two factors greater than 1 (1); composite.	2+1
	OR: LCM$(14, 35, 49) = 490$ min (2); rounds $35, 14, 10$ (1).	2+1
18	Required number $=$ HCF$(3000, 3420)$ (1); HCF$(3000, 3420) = 60$ (2).	1+2
	OR: Take 12 common: $12(10 \times 11 + 1)$ (2); $10 \times 11 + 1 = 111 > 1$, so there are two factors greater than 1 (1); composite.	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Books per stack $=$ HCF$(90, 150, 210) = 30$ (3); stacks $= \frac{450}{30} = 15$ (2).	3+2
	OR: If $1/\sqrt{11} = r$ were rational then $\sqrt{11} = 1/r$ would be rational - contradiction (3); $\frac{2}{\sqrt{11}} = 2 \times \frac{1}{\sqrt{11}}$ is a non-zero rational times an irrational, hence irrational (2).	3+2
20	Books per stack $=$ HCF$(108, 162, 270) = 54$ (3); stacks $= \frac{540}{54} = 10$ (2).	3+2
	OR: If $1/\sqrt{13} = r$ were rational then $\sqrt{13} = 1/r$ would be rational - contradiction (3); $\frac{6}{\sqrt{13}} = 6 \times \frac{1}{\sqrt{13}}$ is a non-zero rational times an irrational, hence irrational (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.